Math 196 Problem Set 2

Due April 9, at 10:00 PM

For book problems, see the pdf on canvas or in my email.

Problem 1. Consider the equation

$$Av = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

(a) If we have

$$A = \begin{bmatrix} -1 & 0 & 2 \\ 1 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix}$$

find a choice of v that makes the equation true. Is it unique?

(b) If we have

$$v = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

find a choice of A that makes the equation true. Is it unique?

Problem 2. For each of the following functions, decide whether or not it is linear. If it is, write down the matrix corresponding to it. If not, show that it is not linear by an example of it failing one of the linearity properties.

$$f \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2y - x \end{bmatrix}$$

$$g(x) = \begin{bmatrix} x \\ 2x + 1 \end{bmatrix}$$

$$h \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} yz \\ xz \\ xy \end{bmatrix}$$

Problem 3. Consider the following matrices.

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 0 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \; C = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}, \; D = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$$

(Note that D is a column vector, but we can also view it as a 3×1 matrix. What linear transformation does it represent? No need to write anything for this.) For each product below, decide whether or not it is well defined and if it is, evaluate it.

$$AB, BA, BC, CB, CD, DC, CBA, CAB$$

Problem 4. Exercises 43 and 44 from Section 2.3 of the book.

Problem 5. Let R_θ and R_φ be the 2×2 matrices representing rotations through angles θ and φ.

(a) Compute the matrix products $R_\theta R_\varphi$ and $R_\varphi R_\theta$, and check that R_φ and R_θ commute.

(b) Reasoning geometrically, we expect that $R_\theta R_\varphi = R_\varphi R_\theta = R_{\theta+\varphi}$. Convince yourself of this. (No need to write anything down.) Using this, derive the trigonometric identities for $\sin(\theta + \varphi)$ and $\cos(\theta + \varphi)$.

Problem 6. In three dimensional space, a rotation about the z-axis through angle θ corresponds to the matrix

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

A rotation about the x-axis through angle θ corresponds to the matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$

Show by example that a rotation around the z-axis and a rotation around the x-axis may not commute.[1]

Problem 7. Suppose f and g are linear maps $\mathbb{R}^n \to \mathbb{R}^m$.

(a) Prove that the map h defined by $h(v) = f(v) + 2 \cdot g(v)$ is also a linear map $\mathbb{R}^n \to \mathbb{R}^m$.

(b) If f corresponds to the matrix A, and g to the matrix B, describe the matrix corresponding to h.

Problem 8. Draw a smiley face in the plane with an eyepatch, centered on the origin. For each transformation below, draw the result of applying the transformation to the face. Note that the last two are given as products of matrices.

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

[1] There are examples where the numbers involved are simple.